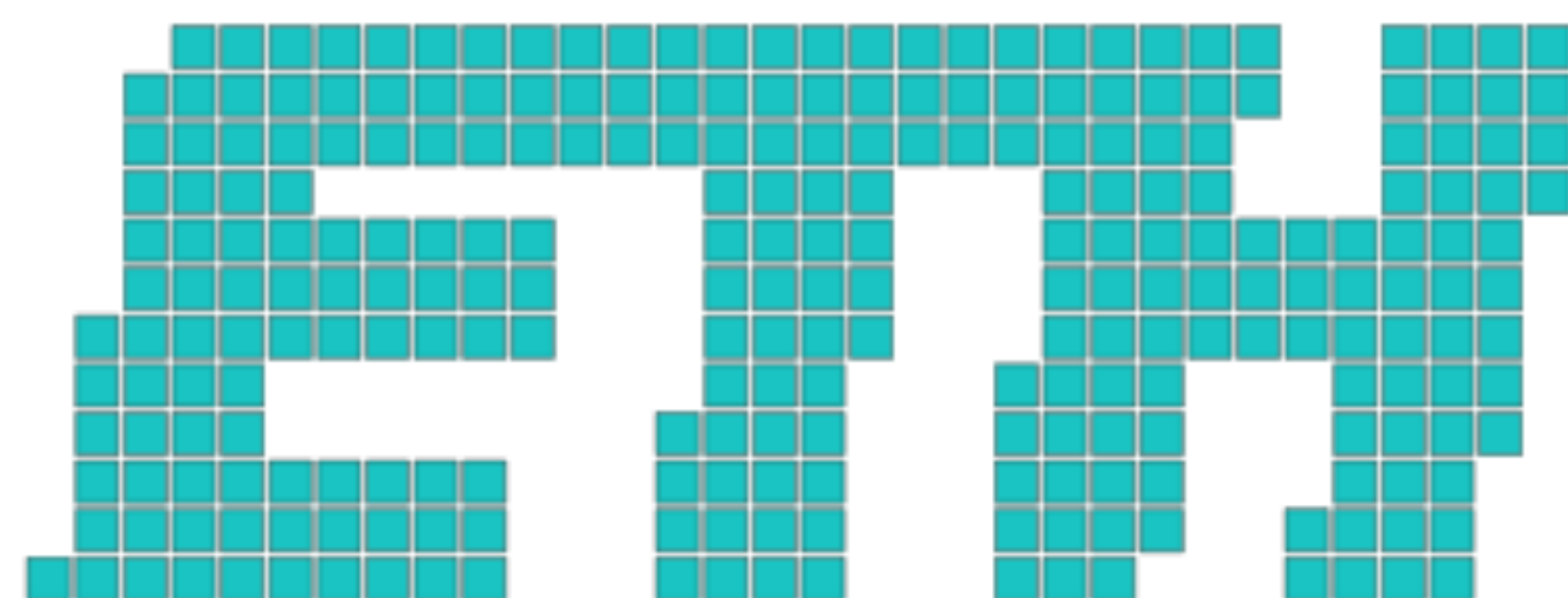
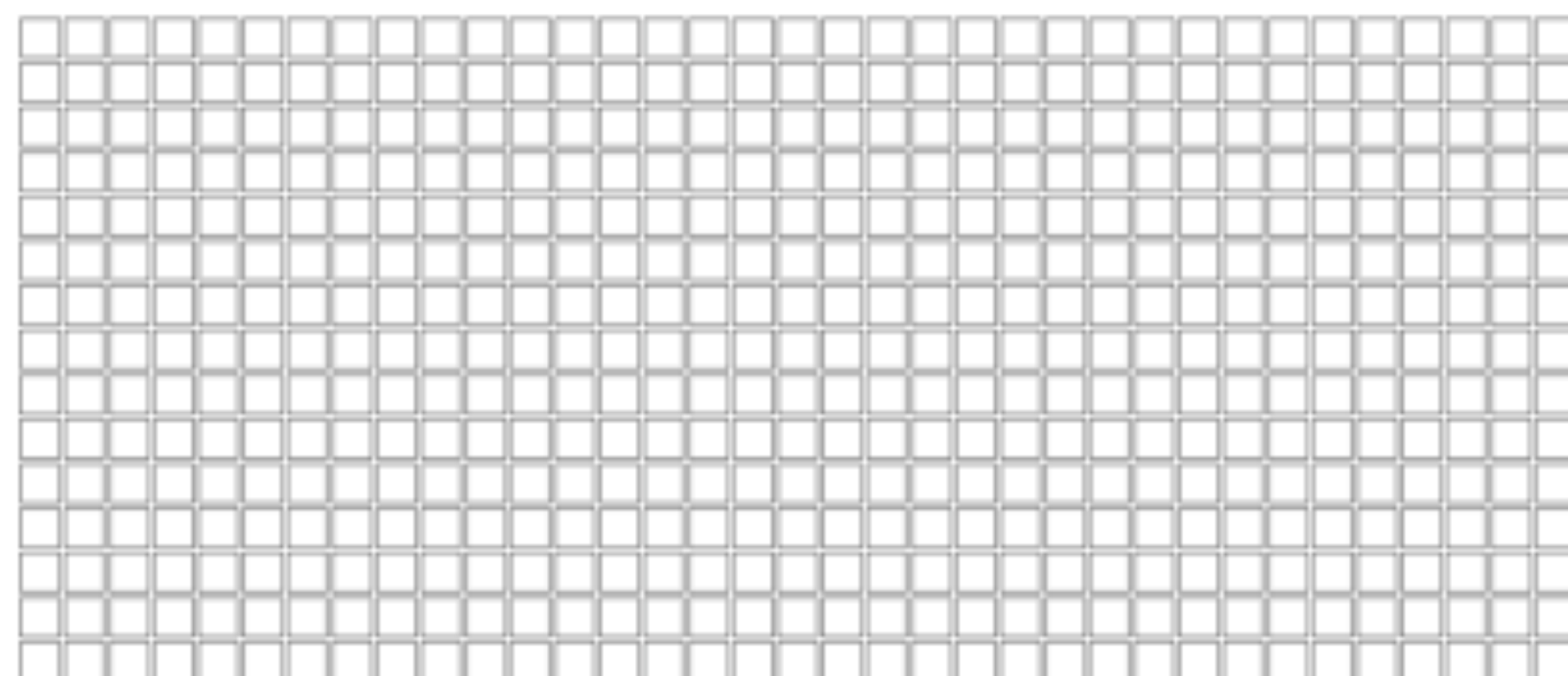




# Complexity Bounds for MCMC with Coupled, Lagged, Unbiased Estimators

FIM Workshop on ‘Scalable MCMC Sampling’, 1 June 2026



Sam Power, University of Bristol  
joint work with Tamás Papp, National University of Singapore



# Integral Estimation by MCMC

# Monte Carlo Basics

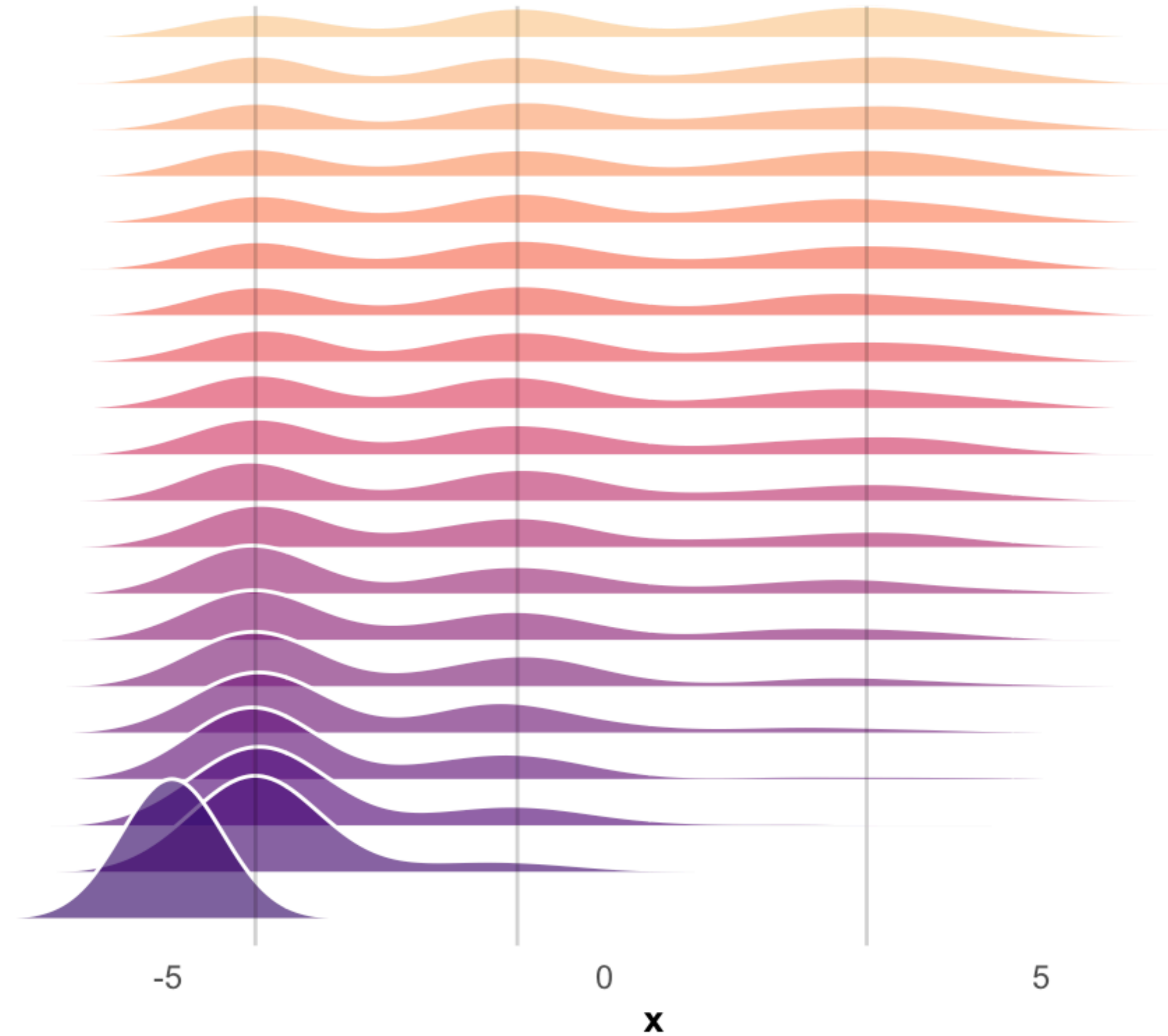
$$\pi(f) \approx \hat{\pi}_N(f) = \frac{1}{N} \sum_{0 < i \leq N} f(X_i)$$

$$X_{-B} \rightarrow \cdots \rightarrow X_0 \rightarrow X_1 \rightarrow \cdots \rightarrow X_L \stackrel{\text{d}}{\approx} \pi$$

$$\pi(f) \approx \hat{\pi}_{B,L}(f) = \frac{1}{L} \sum_{0 < t \leq L} f(X_t)$$

# MCMC Integration

- want that  $\pi(f) \approx \hat{\pi}_{B,L}(f)$  w.h.p.
- variance decays like  $L^{-1}$ ; can estimate
- bias decays like  $\exp(-\Omega(B)) \cdot L^{-1}$ ; harder to estimate

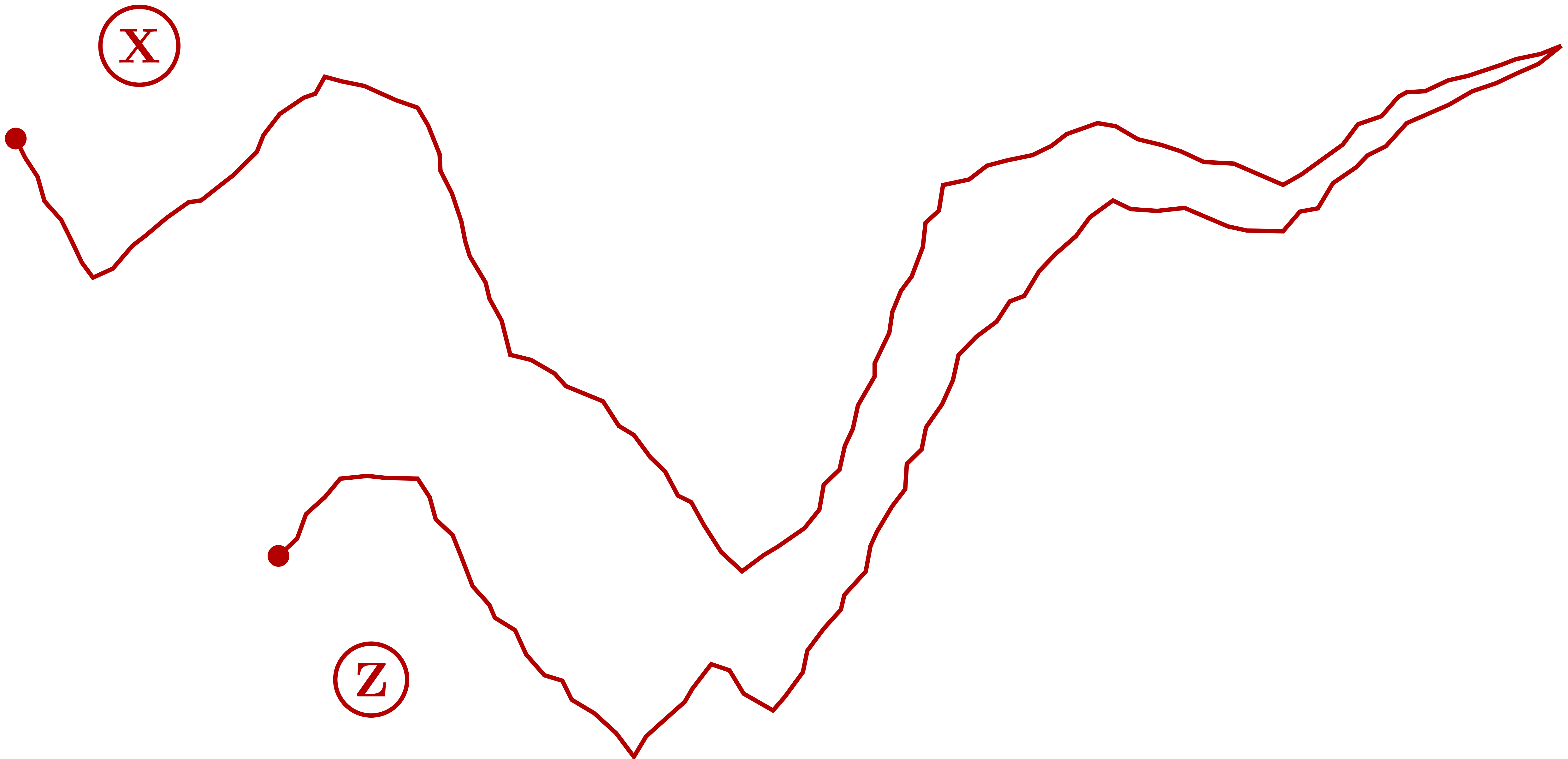


**Are We There Yet?**

# Transient Bias in MCMC

- for ‘good’ Markov chains, bias dissipates quickly (in principle)
- sidestep the issue: remove bias *a priori*
  - *possible?*
- we are making additional demands
  - shall we pay in { *cost, variance, assumptions, ...* } ?
  - shall we pay *heavily*?

# Bias Removal by Coupling



# A Companion Chain

- introduce a second ‘lagged’ Markov chain  $Z$  such that
  - for all  $t$ ,  $Z_t \stackrel{\text{d}}{=} X_{t-L}$ , and
  - as  $t \rightarrow \infty$ ,  $X_t$  and  $Z_t$  move closer together, and then meet
- consider then the estimator

$$\begin{aligned}\hat{\pi}_{B,L,\star}(f) &= \frac{1}{L} \sum_{0 < t \leq L} f(X_t) + \frac{1}{L} \sum_{t > L} \left\{ f(X_t) - f(Z_t) \right\} \\ &= \frac{1}{L} \sum_{0 < t \leq L} f(X_t) + \frac{1}{L} \sum_{L < t < L + \tau_\star} \left\{ f(X_t) - f(Z_t) \right\}\end{aligned}$$



# Coupled Chains in Motion

# An (Unbiased?) Estimator

- under this construction, note that

$$\mathbf{E} \left[ f(X_t) - f(Z_t) \right] = \mathbf{E} \left[ f(X_t) \right] - \mathbf{E} \left[ f(X_{t-L}) \right],$$

and so by a careful telescoping argument, it can be deduced that

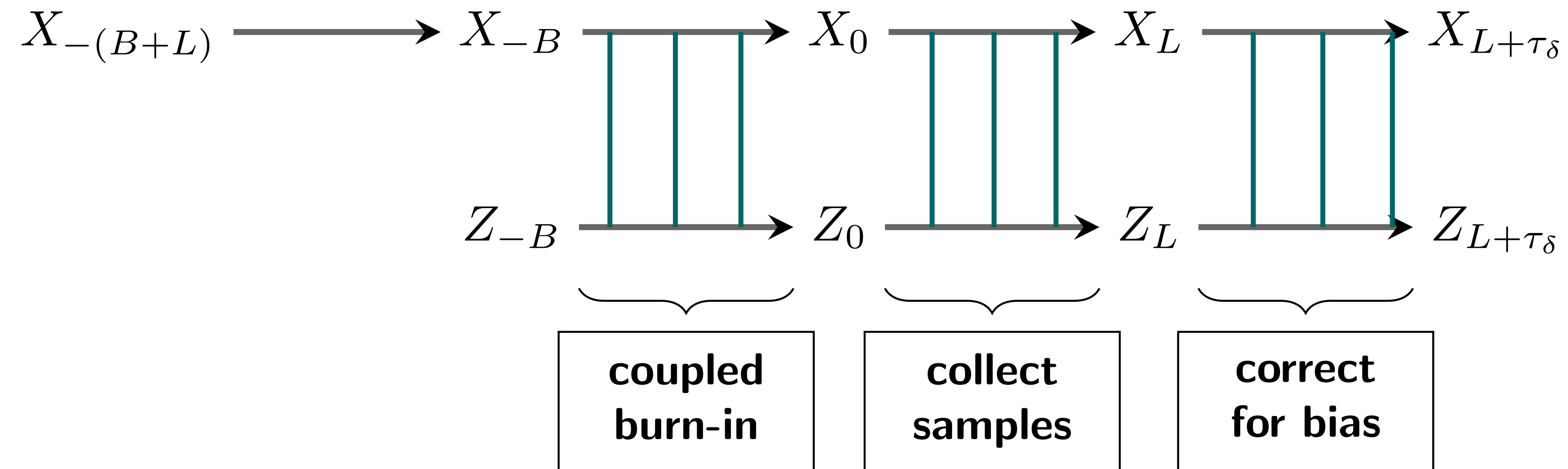
$$\mathbf{E} \left[ \hat{\pi}_{B,L,\star}(f) \right] = \pi(f).$$

- can also terminate upon ‘ $\delta$ -coalescence’ of the chains; ‘ $\delta$ -biased’

# A Method and an Estimator

- initialise  $X_{-B-L} \sim \mu$ 
  - using  $P$ , generate  $\{X_t : -B-L < t \leq B\}$
- initialise  $Z_{-B} \sim \mu$ 
  - using  $\check{P}$ , generate  $\{(X_t, Z_t) : -B < t \leq L\}$
  - using  $\check{P}$ , generate  $\{(X_t, Z_t) : t > L\}$  until  $d(X_t, Z_t) \leq \delta$
- return

$$\hat{\pi}_{B,L,\delta}(f) = \frac{1}{L} \sum_{0 < t \leq L} f(X_t) + \frac{1}{L} \sum_{L < t < L+\tau_\delta} \{f(X_t) - f(Z_t)\}$$



# Coupled, Lagged, Unbiased Estimation



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## Unbiased Markov chain Monte Carlo methods with couplings

Pierre E. Jacob,  
*Harvard University, Cambridge, USA*

John O’Leary  
*Harvard University, Cambridge, and Acadian Asset Management, Boston, USA*

and Yves F. Atchadé  
*Boston University, USA*

[*Read before The Royal Statistical Society at a meeting organized by the Research Section on Wednesday, December 11th, 2019, Professor A. Doucet in the Chair*]

**Summary.** Markov chain Monte Carlo (MCMC) methods provide consistent approximations of integrals as the number of iterations goes to  $\infty$ . MCMC estimators are generally biased after any fixed number of iterations. We propose to remove this bias by using couplings of Markov chains together with a telescopic sum argument of Glynn and Rhee. The resulting unbiased estimators can be computed independently in parallel. We discuss practical couplings for popular MCMC algorithms. We establish the theoretical validity of the estimators proposed and study their efficiency relative to the underlying MCMC algorithms. Finally, we illustrate the performance and limitations of the method on toy examples, on an Ising model around its critical temperature, on a high dimensional variable-selection problem, and on an approximation of the cut distribution arising in Bayesian inference for models made of multiple modules.

**Keywords:** Coupling; Markov chain Monte Carlo methods; Parallel computing; Unbiased estimation

# An Unbiased MCMC Estimator

- this estimator is indeed unbiased, but ...
  - we now have a *random cost*,
  - we have a *different variance*, and
  - we have to *implement a coupling*  $\check{P}$
- the rest: some analysis to elucidate when this methodology *can* work

# Transport Analysis of Markov Chains

# Some Notation, Assumptions

- metric  $d$  induces transport distance  $\mathcal{T} = \mathcal{T}_d$

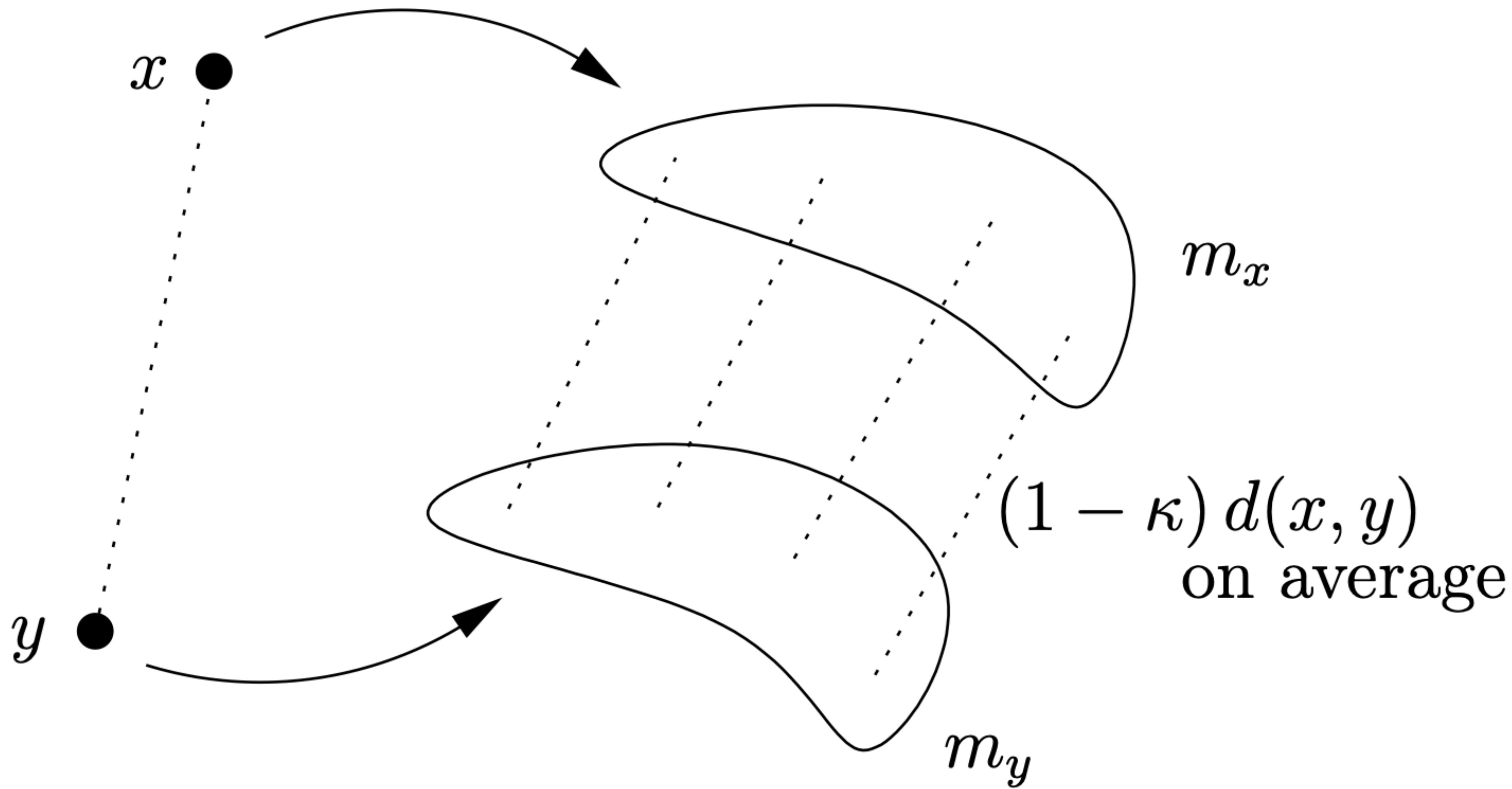
- **curvature:** for all  $x, y$ ,

$$\mathcal{T} \left( P(x, \cdot), P(y, \cdot) \right) \leq (1 - \kappa) \cdot d(x, y)$$

- **local concentration:** for all  $x$ , for all 1-Lipschitz  $f$ ,

$$\mathbf{Var}_{P(x, \cdot)}(f) \leq v$$

- **initialisation:**  $\Delta, \Delta^{(2)}$  denote some sort of initialisation error



# Ricci curvature of Markov chains on metric spaces

Yann Ollivier

*ENS Lyon, UMPA, 46 allée d'Italie, Lyon, France*

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## Abstract

We define the coarse Ricci curvature of metric spaces in terms of how much small balls are closer (in Wasserstein transportation distance) than their centers are. This definition naturally extends to any Markov chain on a metric space. For a Riemannian manifold this gives back, after scaling, the value of Ricci curvature of a tangent vector. Examples of positively curved spaces for this definition include the discrete cube and discrete versions of the Ornstein–Uhlenbeck process. Moreover this generalization is consistent with the Bakry–Émery Ricci curvature for Brownian motion with a drift on a Riemannian manifold.

Positive Ricci curvature is shown to imply a spectral gap, a Lévy–Gromov–like Gaussian concentration theorem and a kind of modified logarithmic Sobolev inequality. The bounds obtained are sharp in a variety of examples.

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*Keywords:* Ricci curvature; Markov chains; Metric geometry; Concentration of measure

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# A Word on Contractivity

- forgivable to think of contractivity as very { specific / strong / ... }
- actually, if you are flexible on the choice of  $d$  ...
  - ... then this is *essentially equivalent* to exponential ergodicity (bonus)

# More on Contractivity

- contraction is our chosen language for ' $P$  mixes well'
- the method is already based on couplings: *natural*
- assume today that our algorithmic coupling  $\check{P}$  'contracts optimally'

# Integration under Contractivity

# **CURVATURE, CONCENTRATION AND ERROR ESTIMATES FOR MARKOV CHAIN MONTE CARLO**

BY ALDÉRIC JOULIN AND YANN OLLIVIER

*Université de Toulouse and CNRS and École normale supérieure de Lyon*

We provide explicit nonasymptotic estimates for the rate of convergence of empirical means of Markov chains, together with a Gaussian or exponential control on the deviations of empirical means. These estimates hold under a “positive curvature” assumption expressing a kind of metric ergodicity, which generalizes the Ricci curvature from differential geometry and, on finite graphs, amounts to contraction under path coupling.

***“Some even say that confidence intervals disappeared the day MCMC methods appeared”***

# Analysis of MCMC (1)

- Joulin-Ollivier provide bias and variance bounds for ‘usual’ MCMC
- proof by martingale / Duhamel decomposition
  - dynamics are well-concentrated, hence small incremental variance
  - dynamics are contractive, hence dissipate errors

# Analysis of MCMC (2)

- with  $\sigma^2 = v \cdot \kappa^{-2}$ ,  $f$  Lipschitz, obtain

$$\text{bias} \leq \frac{(1 - \kappa)^B}{\kappa L} \cdot \Delta$$

$$L \cdot \text{variance} \leq \sigma^2 + \frac{\sigma^2}{\kappa L} + \frac{(1 - \kappa)^{2B}}{\kappa^2 L} \cdot \Delta^{(2)}$$

- obtain  $\text{MSE} \leq \epsilon^2$  by taking  $\kappa B \gtrsim \log \left( \frac{\Delta}{\epsilon} \right)$ ,  $\kappa L \gtrsim \frac{\sigma^2}{\epsilon^2}$

# Analysis of CLUE (1)

- the CLUE is expressible as a ‘correction’ of the usual MCMC estimator
- for variance, sufficient to control second moment of this correction

$$\mathbf{E} \left[ \left| \frac{1}{L} \sum_{L < t < L + \tau_\delta} \left\{ f(X_t) - f(Z_t) \right\} \right|^2 \right] \leq ?$$

- for  $\delta > 0$ , also still need to assess bias

# Analysis of CLUE (2)

- with ‘ $\delta$ -coalescence’ construction, obtain

$$\text{bias} \leq \frac{\delta}{\kappa L}$$

$$|\text{correction}|^2 \leq \frac{(1 - \kappa)^{2(B+L)}}{\kappa^2} \cdot \Delta^{(2)}$$

$$\text{cost}_{1-\alpha} \leq \max \left\{ B + L, L + \frac{1}{\kappa} \cdot \log \left( \frac{\Delta^{(2)}}{\alpha \delta} \right) \right\}$$

# Analysis of CLUE (3)

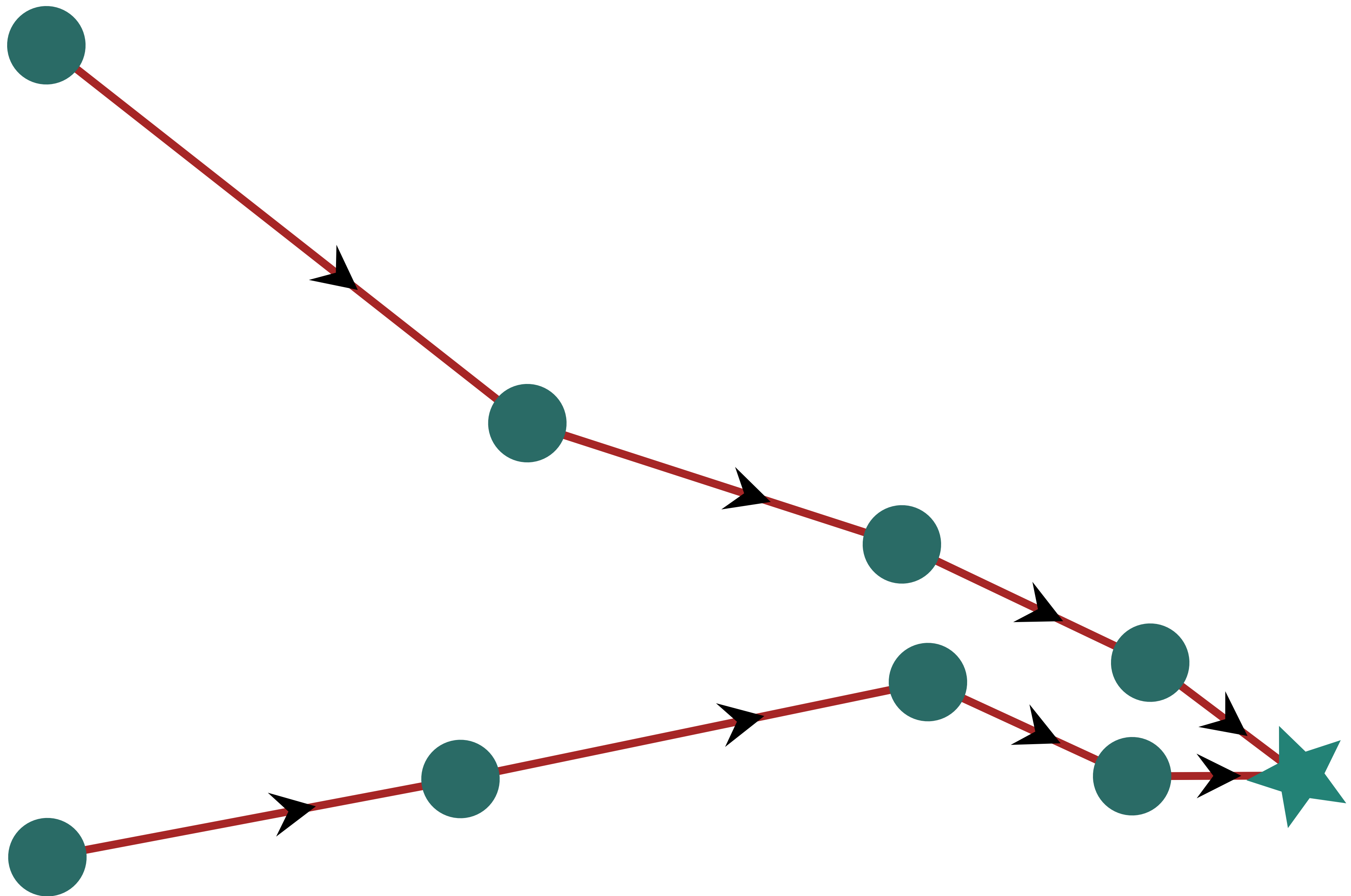
- to control  $\text{MSE} \leq \epsilon^2$ , suffices to take

$$\kappa B \gtrsim \log \left( \frac{\max \left\{ \Delta, \sqrt{\Delta^{(2)}} \right\}}{\epsilon} \right), \quad \kappa L \gtrsim \frac{\sigma^2}{\epsilon^2}$$

- comparable to usual MCMC
  - underwhelming (?), but this is the point

# Exact Coalescence?

- easiest to describe with  $\delta > 0$  ...
  - ... but usually implemented with  $\delta = 0$ 
    - (if you *truly* want to be ‘unbiased’)
- typical intuition: contract spatially until close, then try to meet exactly



# Coalescence and Hybrid Metrics

- perspective: you are contracting some *hybrid* metric  $d$  ...
  - ... which satisfies e.g.  $d(x, y) \geq 1 [x \neq y]$  ...
    - ... so, just run analysis with  $\delta = 1$ !

# Summary and Outlook

# Takeaways

- *transient bias* in MCMC can be systematically removed by **coupling**
- only a *modest inflation* of cost and variance
- in contrast to MCMC, need a good coupling in *principle* and in *practice*
- variance bounds give indications on how to *allocate computational effort*
  - more practical recommendations in the pipeline

**- - - Bonus Material - - -**

# A Universal Metric for Curvature

- consider

$$d_{\rho}(x, y) = \inf \left\{ \sum_{t \geq 0} \frac{d(X_t, Y_t)}{\rho^t} : \text{path-space couplings} \right\}$$

- if finite, then contracts at rate  $\rho$ 
  - ‘usually’ finite for  $\rho \in (\rho_{\star}, 1)$

# Unbiasedness of CLUE (1)

$$\begin{aligned}\hat{\pi}_{B,L,\star}(f) &= \frac{1}{L} \sum_{0 < t \leq L} f(X_t) + \frac{1}{L} \sum_{t > L} \left\{ f(X_t) - f(Z_t) \right\} \\ &= \lim_{T \rightarrow \infty} \left\{ \frac{1}{L} \sum_{0 < t \leq L} f(X_t) + \frac{1}{L} \sum_{L < t < L+T} \left\{ f(X_t) - f(Z_t) \right\} \right\}\end{aligned}$$

# Unbiasedness of CLUE (2)

$$\begin{aligned}\mathbf{E} \left[ \hat{\pi}_{B,L,\star;T}(f) \right] &= \frac{1}{L} \sum_{0 < t \leq L} \mathbf{E} \left[ f(X_t) \right] + \frac{1}{L} \sum_{L < t < L+T} \mathbf{E} \left[ f(X_t) - f(Z_t) \right] \\ &= \frac{1}{L} \sum_{0 < t \leq L} \mathbf{E} \left[ f(X_t) \right] + \frac{1}{L} \sum_{L < t < L+T} \mathbf{E} \left[ f(X_t) - f(X_{t-L}) \right] \\ &= \frac{1}{L} \sum_{0 < t \leq L+T} \mathbf{E} \left[ f(X_t) \right] - \frac{1}{L} \sum_{0 < t \leq L} \mathbf{E} \left[ f(X_t) \right] \\ &= \frac{1}{L} \sum_{L < t \leq L+T} \mathbf{E} \left[ f(X_t) \right] \xrightarrow[T \rightarrow \infty]{} \pi(f)\end{aligned}$$



**Are We There Yet?**

